Biochemistry Cumulative Examination
February 29, 2008

Protein production and purification is critical to many biochemical research products. This area was reviewed by a variety of structural genomics groups engaged in attempts to produce in milligram quantities tens of thousands of proteins. Based on your reading of the article in *Nature Methods* (2008, **5**, 135-146) and your general understanding of biochemistry, answer the following questions.

1. One topic discussed is codon bias. Briefly explain what codon bias is and how it can affect protein production. (20 points)

2. Purification is often by IMAC. What does this acronym stand for and explain briefly how it works. (20 points)

3. The most common vectors used are the pET family. What promoter is used in this vector and why is it favored? (20 points)

4. It is suggested that if unable to produce protein on the first attempt, the first troubleshooting step should be to lower the temperature of growth. What does this do and why does it help? (20 points)

5. It is also suggested that making several different constructs with differing N and C termini is useful. Why? (20 points)